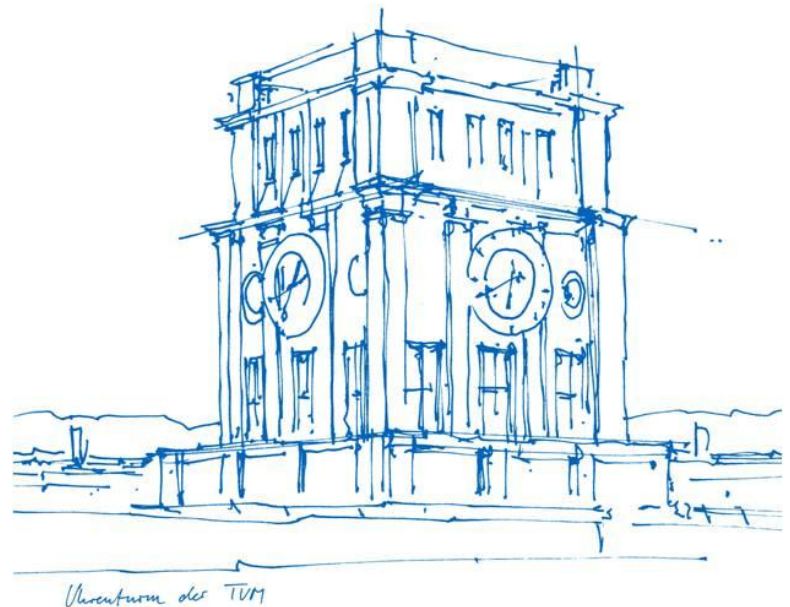


Real Time SSL and Tracking System

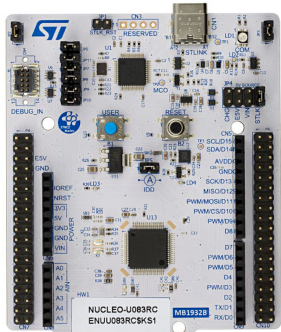
Presented by Samuele Ribaudo and Hong Yan Jun



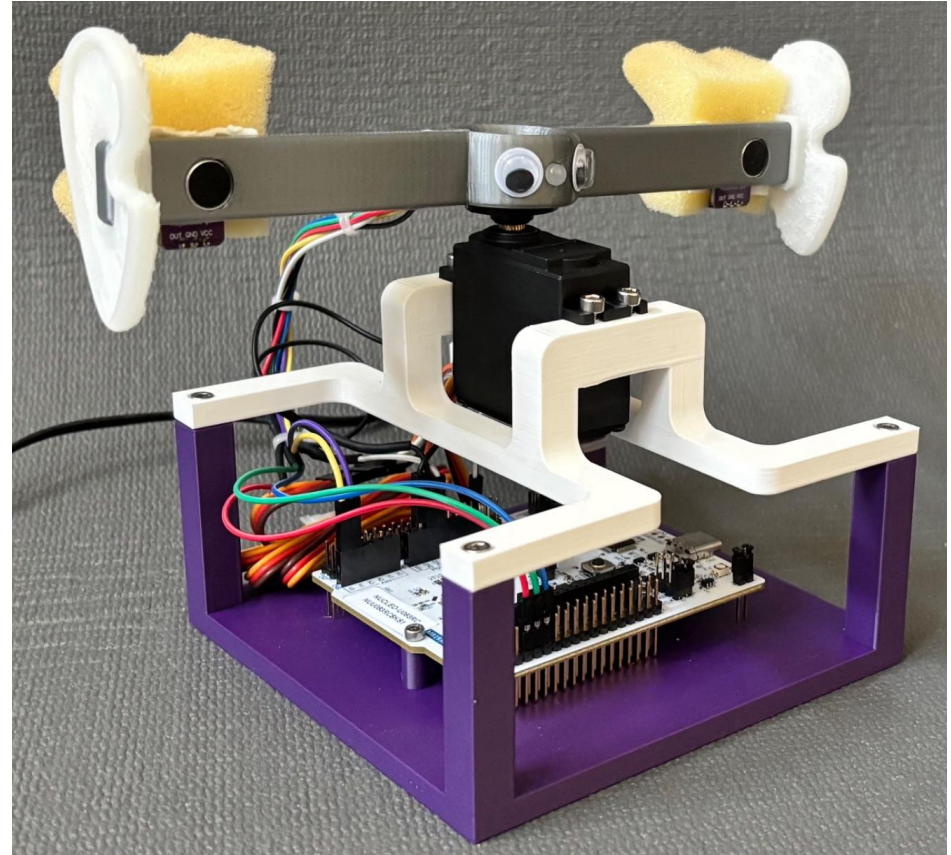
Introduction

Robotic “hearing system”

- Two analog microphones.
- Compute angle from which human voice is coming from.
- Physically rotate the servo motor.
- Modularity → onboard RGB LED

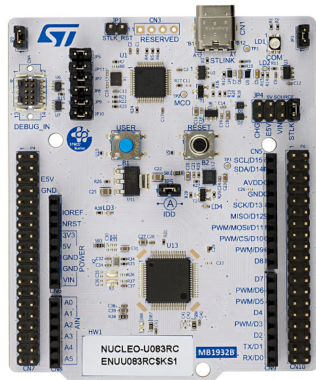


STM32 Nucleo
microcontroller



MCU configuration & application logic

First challenge

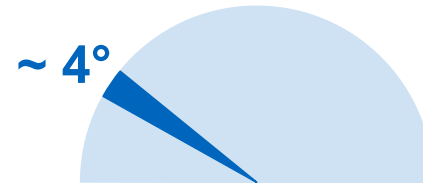


⚠ Limited memory

→ 50 kHz →



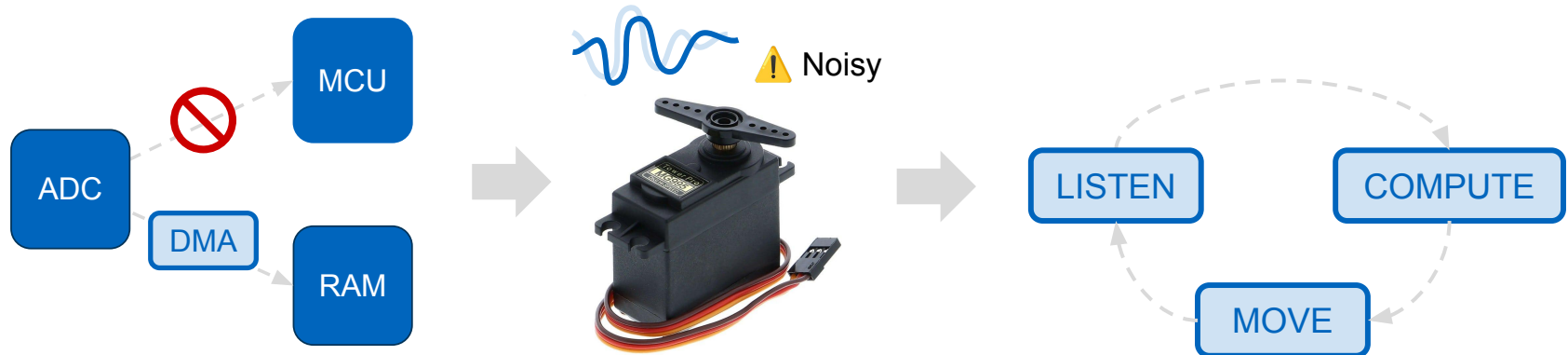
~ 20 ms



~ 4°

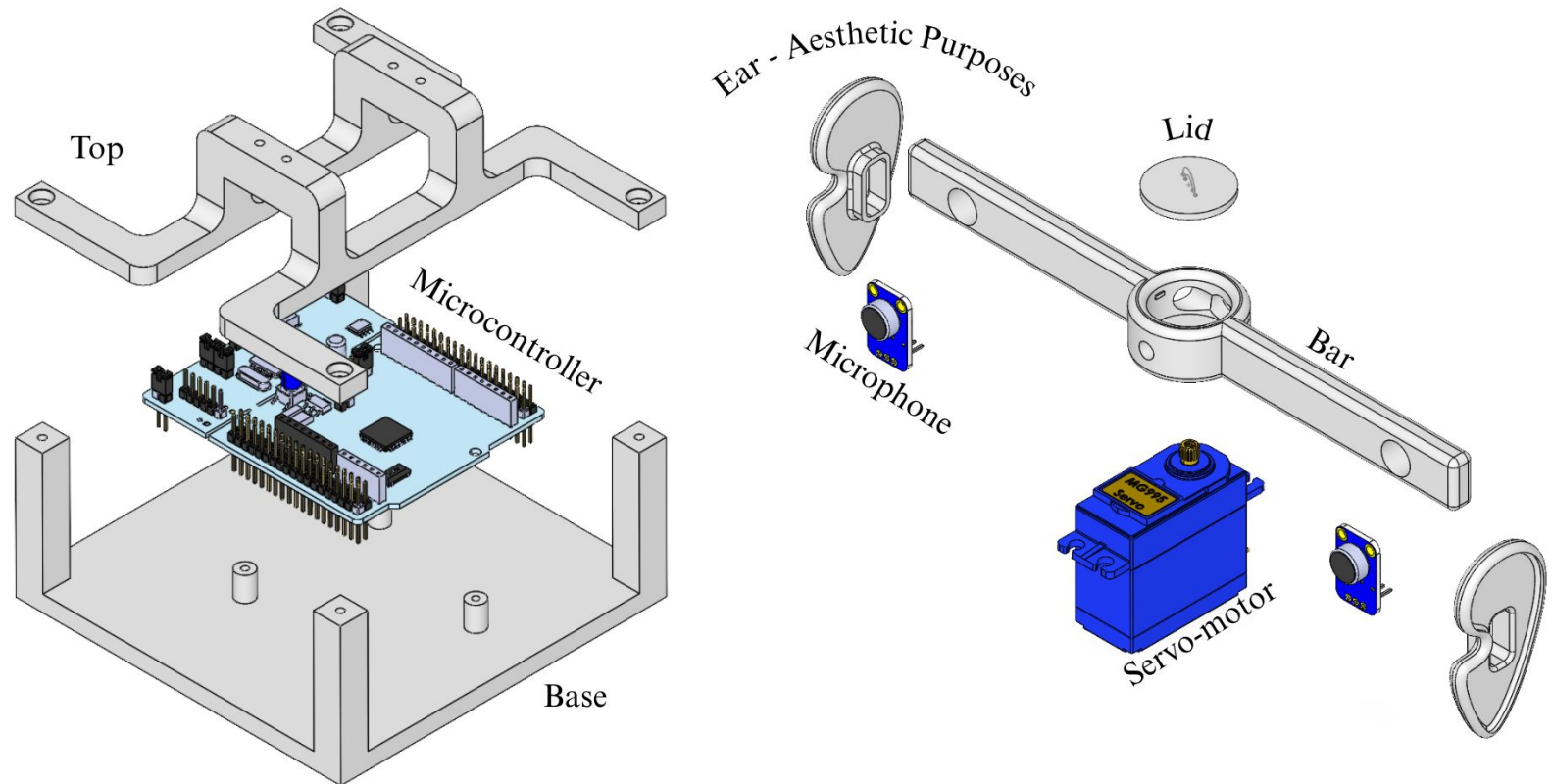
MCU configuration & application logic

Need for a FSM



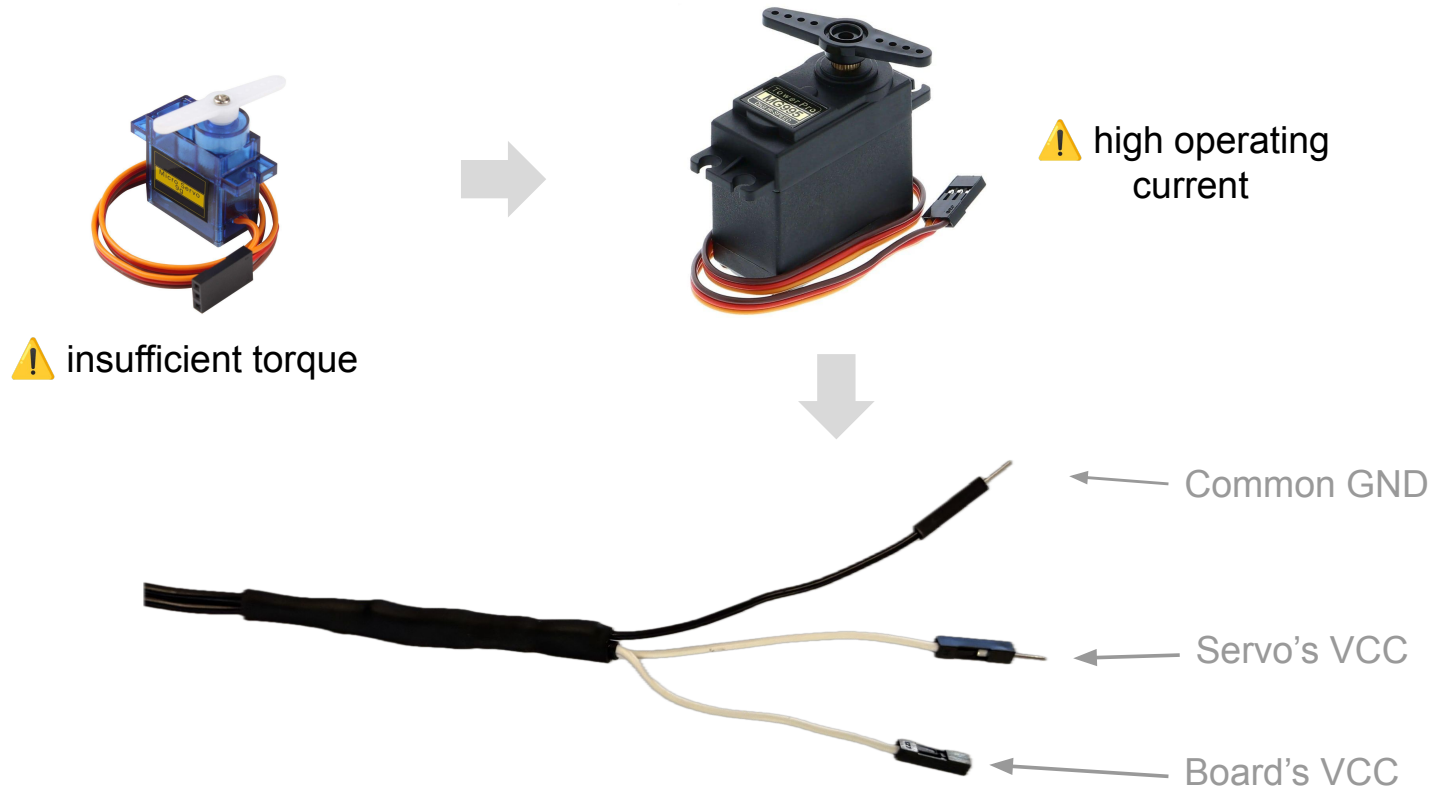
CAD design & hardware fabrication

The 3D design



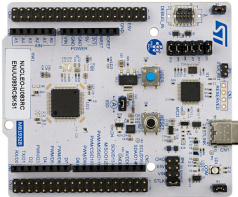
The servo challenge

- **Problem:** Servo motor shaking violently around center position
- **Solution:**



DSP pipeline

Signal filtering



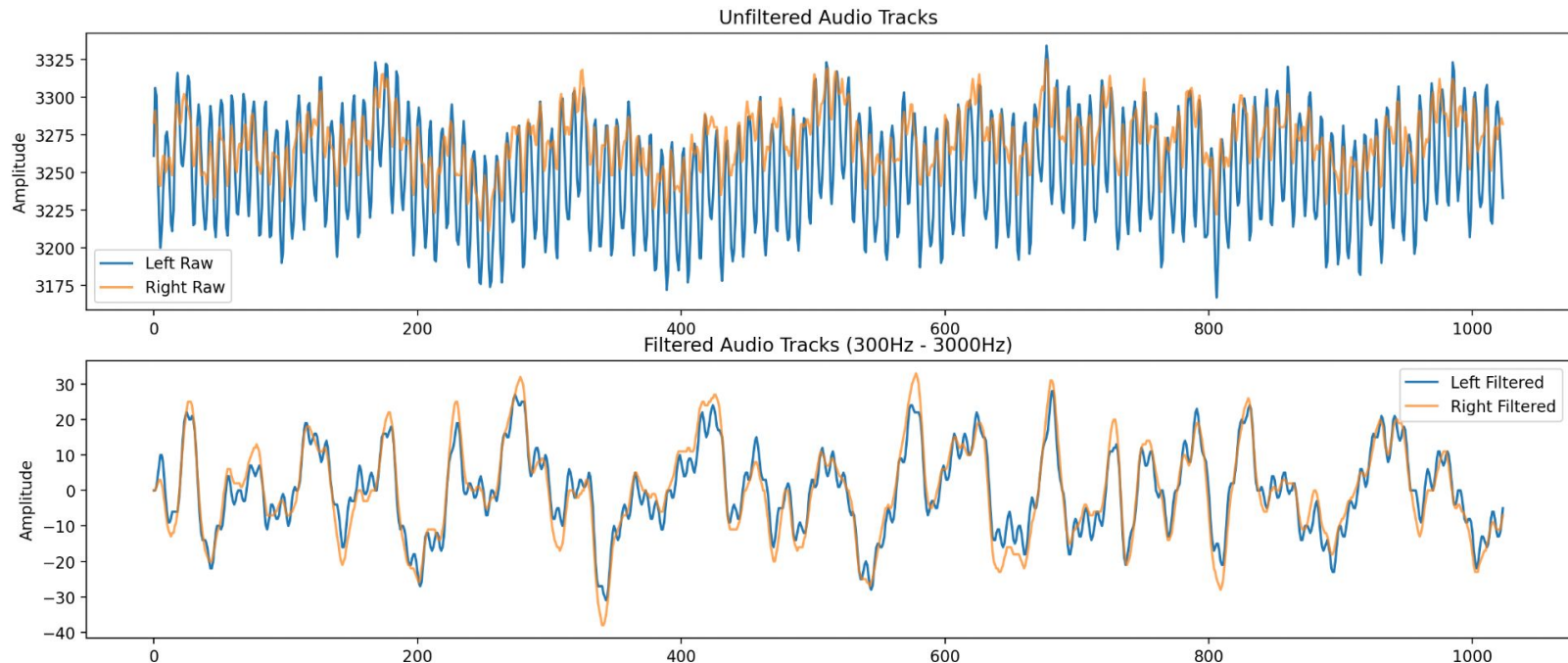
Custom
DSP
pipeline

- 6th order Butterworth bandpass filter

300 Hz - 3000 Hz



⚠ Limited memory

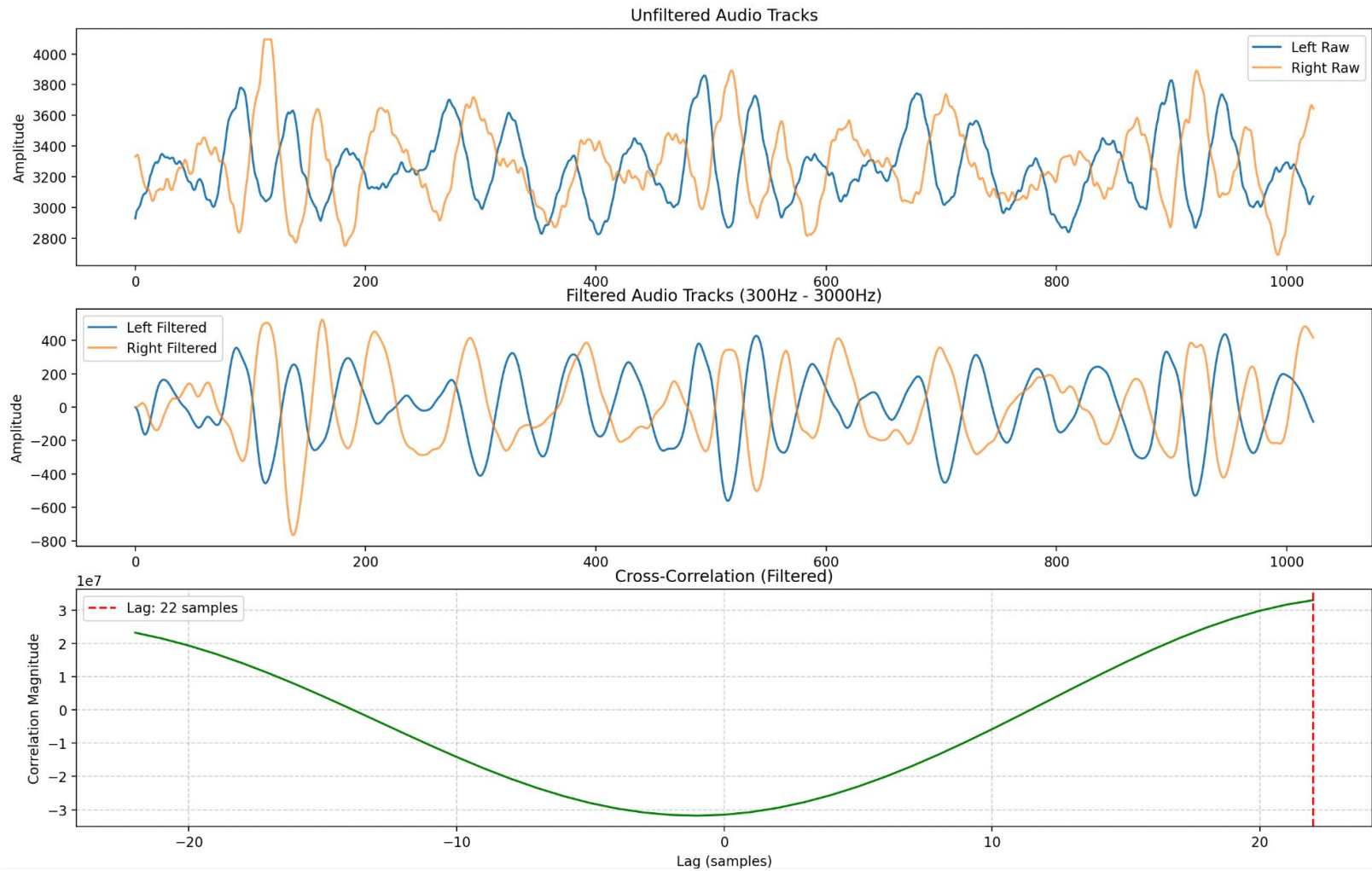


Signal processing

- To remove the DC offset - Mean-centering
- To find the Time Delay of Arrival - Cross-correlation
 - Problem: 1024- sample arrays too much for our MCU
 - Solution
 - Knowing speed of around and distance between microphone
 - Bound search loop to maximum delay of ± 22 samples

DSP pipeline

Signal processing



Conclusions

QR codes

Embedded pipeline successfully localizes sound sources in 1D-space in real time, ignoring noises outside human speech frequencies.

